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MINI-REVIEW

Methanogens in humans: potentially beneficial or harmful for health

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Abstract
Methanogens are anaerobic prokaryotes from the domain archaea that utilize hydrogen to reduce carbon dioxide, acetate, and a variety of methyl compounds into methane. Earlier believed to inhabit only the extreme environments, these organisms are now reported to be found in various environments including mesophilic habitats and the human body. The biological significance of methanogens for humans has been re-evaluated in the last few decades. Their contribution towards pathogenicity has received much less attention than their bacterial counterparts. In humans, methanogens have been studied in the gastrointestinal tract, mouth, and vagina, and considerable focus has shifted towards elucidating their possible role in the progression of disease conditions in humans. Methanoarchaea are also part of the human skin microbiome and proposed to play a role in ammonia turnover. Compared to hundreds of different bacterial species, the human body harbors only a handful of methanogen species represented by *Methanobrevibacter smithii*, *Methanobrevibacter oralis*, *Methanosphaera stadtmanae*, *Methanomassiliicoccus luminyensis*, *Candidatus* Methanomassiliicoccus intestinalis, and *Candidatus* Methanomethylophilus alvus. Their presence in the human gut suggests an indirect correlation with severe diseases of the colon. In this review, we examine the current knowledge about the methanoarchaea in the human body and possible beneficial or less favorable interactions.

Keywords Archaea · Methanogens · Human gut · Immune response · Microbiome

Introduction

Archaea are prokaryotic microorganisms generally thriving in the most extreme, harsh, and inhabitable environments. These extremophiles represent the third domain of life and are different from bacteria in their 16S rRNA sequences, metabolic characteristics, transcription, and translation mechanisms but share some characteristics with eukaryotes (Woese and Fox 1977; Woese et al. 1990). Methanogens are involved in hydrogenotrophic metabolism requiring the presence of H_2 to reduce CO_2 to methane, a process termed as methanogenesis (Gaci et al. 2014). Alternatively, methanogens can convert acetate into CH_4 (acetoclastic pathway) or methanol/methylamines into CH_4 (methylotrophic pathway). Methanogenic archaea are extremely oxygen-sensitive and free radical-sensitive prokaryotic organisms that survive only in a strongly reduced environment having a redox potential < −200 mV (Karakashev et al. 2006).

Extensive studies in the last three decades have now confirmed their existence in some familiar mesophilic habitats including oceans, soils, river sediments, sewage, marshlands, paddy fields, gastrointestinal tract of animals, rumen, human intestine, skin, and vagina. (Delong 1998; Chaudhary et al. 2012, 2014; Buriánková et al. 2013; Brablcová et al. 2015; Mach et al. 2015). Molecular diversity of methanogens is determined on the basis of *mcrA* or 16S rRNA gene sequences (Sirohi et al. 2013). The *mcrA* gene encodes a portion of methyl-coenzyme reductase (Methyl-CoM) that is critical for the completion of the methanogenesis pathway (Sirohi et al. 2013). The *mtbA*, *mtaB*, *cdhD1*, and *mer* genes have been utilized in various studies for monitoring the specific pathway in acetoclastic and methylotrophic methanogens (Borrel et al. 2013). Using T-RFLP and DDGE molecular profiling, it has

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been shown that in river sediments, the dominant pathway of methanogenesis is hydrogenotrophic (Chaudhary et al. 2014; Mach et al. 2015). The dominant orders of methanogens were *Methanosarcinales*, *Methanobacteriales*, and *Methanomicrobiales* (Mach et al. 2015; Chaudhary et al. 2017). Recently, *mcrA* gene libraries have been used to produce a more accurate look at the population dynamics of methanogens in ruminants (Chaudhary and Sirohi 2009; Chaudhary et al. 2012; Sirohi et al. 2013). Using the *mcr*A gene, it has been determined that the dominant order of methanogens in ruminants is *Methanobacterials* and specifically the *Methanobrevibacter* species. Other major orders of methanogens in ruminants include *Methanosarcinales* and *Methanomicrobiales* (Sirohi et al. 2013). It is therefore possible to apply the methods used in these environments to the human gastrointestinal tract to help better understand methanogenic population dynamics in humans.

Methanogens in the human gastrointestinal tract

The last decade witnessed renewed interest towards the contribution of methanogens in human disease with the application of molecular and improved culture techniques. Methanogens (or methanogenic archaea) are found in the human gastrointestinal tract. Microbial interactions with the host are known to exert physiological and biochemical effects which can be either beneficial or detrimental (Conway de Macario and Macario 2009). Archaea are typical representatives of the Healthy Mature Anaerobic Gut Microbiota (HMAGM) (Million et al. 2016), and it has been shown that they are part of the developing microbiota of the infant (Wampach et al. 2017). It has been estimated that about 33% of the world's human population is considered to be methane producers (Keppler et al. 2016).

In a healthy individual, there are extremely low levels of oxygen and over 90% of gut microbiota is represented by strict anaerobes. For adult Europeans, about 0.8% of the fecal microbiota genes have been assigned to archaea (Qin et al. 2010), and similar numbers (0.2–0.3%) were reported for Amerindians and Malawians (Yatsunenko et al. 2012), while North Americans had lower fractions of archaeal genes i.e. less than 0.05% (Lurie-Weinberger and Gophna 2015).

Miller et al. (1982) were the first to report methanogens in human when they isolated *Methanobrevibacter smithii* from intestinal contents (Miller et al. 1982). *Methanobrevibacter smithii* was found to utilize H_2 and CO_2 for methanogenesis. Subsequently, Miller and Wolin discovered *Methanosphaera stadtmanae* which was utilizing methanol and H_2 for metabolism (Miller and Wolin 1983). In 2008, *mcrA* gene sequence-based studies identified a new order named *Methanomassiliicoccales* of methanogens in the gut (Mihajlovski et al. 2008). The human gut harbors three representatives of this order, namely, *Methanomassiliicoccus luminyensis*, *Candidatus* Methanomethylophilus alvus, and *Candidatus* Methanomassiliicoccus intestinalis (Borrel et al. 2012; Borrel et al. 2013). Members of this order require H_2 to perform methanogenesis just like *Methanobrevibacter smithii* and *Methanosphaera stadtmanae*; however, rather than reducing CO_2, they utilize the methyl group of methanol and methylamines as substrate (Brugère et al. 2014). So far, no acetoclastic methanogens have been detected in humans.

Breath methane content can be used to detect methanogens. Studies have revealed that 33–36% of the microbiota can be methane producers (Miller and Wolin 1983; Levitt et al. 2006). Although methanogen populations are believed to undergo successive dynamic changes throughout an individual's life, it is suggested that these populations remain reasonably stable in adults (Abell et al. 2006; Hansen et al. 2011).

It has been reported that the impact of methanogens of the gut microbiota on pathological conditions has been grossly underestimated (Bang et al. 2014). *Methanobrevibacter smithii* and *Methanosphaera stadtmanae* are specifically recognized by the human innate immune system, and it is considered that these species contribute to immune responses in the body, including inflammatory responses (Bang et al. 2014). Although the archaea of the human colon are generally always methanogens, there is one report of halophilic archaea found in biopsies from inflammatory bowel disease patients (Oxley et al. 2010). Most of these strict anaerobes belong to the order *Methanobacteriales* with *Methanobrevibacter* and *Methanosphaera* being the dominating representative genera (Lurie-Weinberger and Gophna 2015).

Methanobrevibacter was first isolated from a human stool sample in 1968 and named *Methanobacterium* (Nottingham and Hungate 1968). Later, it was confirmed that these fecal isolates were members of *Methanobrevibacter smithii* (Miller et al. 1982). This species is reported to be the most abundant methanogen in the human body, thus accounting for about 10% of all anaerobes found in the colon (Eckburg et al. 2005; Samuel et al. 2007). It is reported to be present in approximately 95% of human subjects (Dridi et al. 2009). In the human gut, methanogenesis substrates (H_2, methanol, and acetate) are mostly obtained from bacterial fermentation. It has been reported that the abundance of *Methanobrevibacter smithii* appears to remain stable over time, even following radical dietary changes (Miller and Wolin 1983). It is suggested to be highly heritable because of its presence in monozygotic twins compared to dizygotic twins (Hansen et al. 2011; Goodrich et al. 2014).

The second most abundant methanogen present in gut is *Methanosphaera stadtmanae*, which has a highly restricted energy metabolism since it is totally dependent on methanol as its sole carbon source and utilizes methanol and H_2 for methane production (Miller and Wolin 1985). Recently, it has been shown that both of these species induce monocyte-derived dendritic cell maturation. *Methanosphaera stadtmanae* induced a pro-inflammatory cytokine release from monocyte-derived dendritic cells (Bang et al. 2014) and is prevalent in inflammatory bowel disease patients (Lecours et al. 2014). An overview of the different types of diseased conditions which have been associated with the counts of methanogens present in the human gut and oral cavity is presented in Fig. 1.

Archaea on the human skin

There have been some reports of strict anaerobic archaea on human skin (Probst et al. 2013; Lurie-Weinberger and Gophna 2015). A group of researchers detected *Methanobrevibacter* on the skin of several individuals using 16S rRNA gene sequences and confirmed the presence of archaea on the skin of 13/13 volunteers, with relative abundances that exceeded 4% in one individual (Probst et al. 2013). These workers showed that skin phylotypes encode *amoA* gene homologs and are therefore likely to be chemolithotrophic ammonium oxidizers (Probst et al. 2013). Sweating and exercise will lead to increased excretion of ammonia on the skin so the individuals who sweat more or exercise more are more likely to have archaeal communities present on their skin (Sato et al. 1989; Czarnowski and Górski 1991). Skin-associated archaea belonging to the third archaeal phylum *Thaumarchaeota* were found on left and right palms of one male and one female subject over several months (Caporaso et al. 2011). Phylotypes belonging to the halophilic archaea from family *Halobacteriaceae* were obtained from an individual who abstained from showers or baths for several years (Hulcr et al. 2012).

Subgingival archaea

Three genera of methanogens, namely, *Methanobrevibacter*, *Methanosphaera*, and *Methanosarcina*, have been isolated from subgingival dental plaque; however, there is a low level of diversity of archaea in human subgingival dental plaques. *Methanobrevibacter oralis* is by far the most prevalent methanogen found in this environment (Table 1). *Methanobrevibacter oralis* is significantly associated with periodontal disease and is considered a likely pathogen (Nguyen-Hieu et al. 2013). Methanogens indirectly support the development of periodontal disease through syntrophic interactions with sulfate-reducing bacteria as revealed by studies in an animal model of periodontal disease. Metronidazole is highly effective against *Methanobrevibacter oralis* and is commonly used to treat periodontitis. Statins which inhibit the activity of 3-hydroxy-3-methylglutaryl coenzyme A

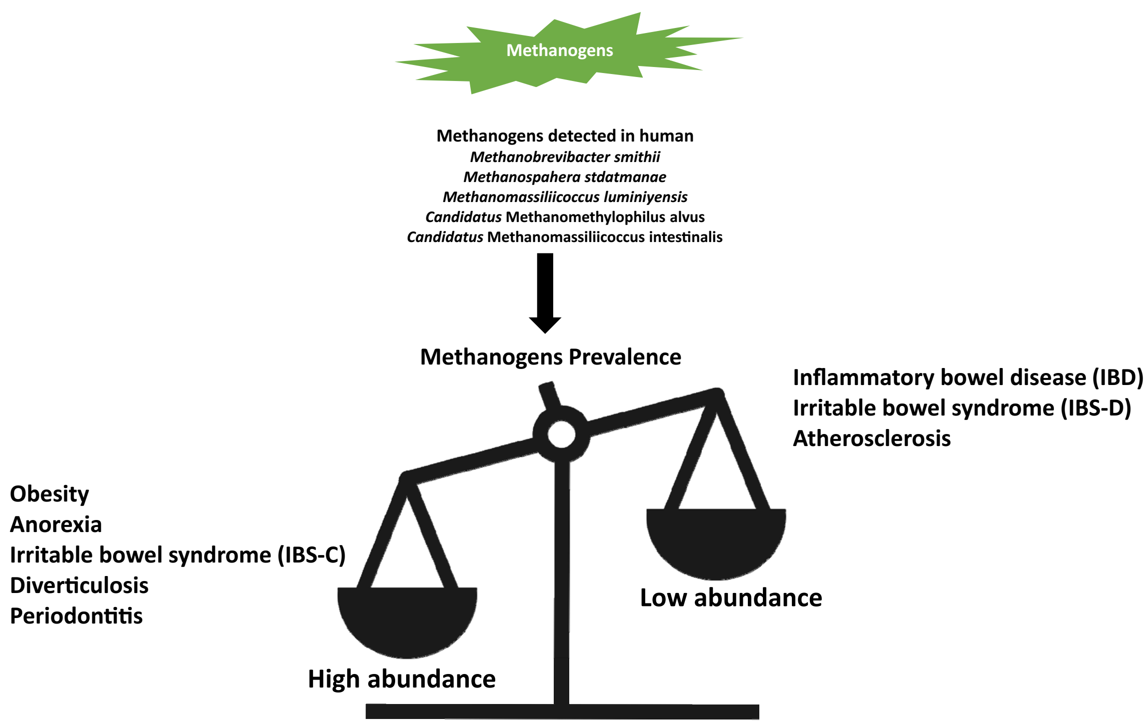

Fig. 1 Overview of the different types of diseased conditions which have been associated with the amount of methanogens present in the human gut and oral cavity as compared to the control subjects in different studies. In IBS-C and IBS-D, C and D refer to constipation and diarrhea, respectively. Note: No clear evidence or study reports the relationship of methanogens levels with colorectal cancer

Table 1 Methanogenic species associated with in vivo methanogenesis and pathological conditions in and on the human body

Methanoarchaea	Order	Habitat	Diversity	Disease conditions/ health ailments	Clinical developments	Reference
Methanobrevibacter smithii	*Methanobacteriales*	GI tract, oral cavity, vagina	94% of the methanogen population	Constipation, irritable bowel syndrome with constipation (IBS-C), and obesity	A phase 2 clinical trial is currently in progress that evaluates the effect of lovastatin lactone on methanogenesis and symptoms in patients with irritable bowel syndrome with constipation.	Miller et al. 1982; Pimentel et al. 2012; Gottlieb et al. 2016
Methanobrevibacter oralis	*Methanobacteriales*	Subgingival dental plaque; periodontitis-related lesions and peri-implant pockets		Periodontal disease by syntrophic interactions with sulfate-reducing bacteria		Ferrari et al. 1994; Nguyen-Hieu et al. 2013
Methanosphaera stadtmanae	*Methanobacteriales*	Inflammatory bowel disease		Prevalence of MSS is increased in IBD patients and is associated with an antigen-specific IgG response		Lecours et al. 2014
Methanomassiliicoccus luminyensis	*Methanomassiliicoccales*	GI tract				Dridi et al. 2012; Iino et al. 2013
Candidatus Methanomethylophilus alvus	*Methanomassiliicoccales*	GI tract				Borrel et al. 2012
Candidatus Methanomassiliicoccus intestinalis	*Methanomassiliicoccales*	GI tract				Borrel et al. 2013
Candidatus Methanobrevibacter sp. N13	*Methanobacteriales*			Periodontitis lesions		Huynh et al. 2015; Huynh et al. 2016

reductases of archaeal could also be exploited as therapeutic drug to treat periodontal disease.

Role of methanogens in human diseased conditions

The issue as to whether methanogens cause, support, or prevent diseases and health conditions in humans is debated. The crucial question whether methanogens are pathogenic or beneficial still remains far from being fully answered. There is evidence that *M. smithii* assists the growth of fermentative bacteria in the gut by decreasing H_2 levels, thus establishing a syntrophic association (Samuel and Gordon 2006). It is assumed that in malnourished individuals, *M. smithii* would increase caloric uptake and prove beneficial. It is essential to have a sound understanding of methanogen population dynamics to devise effective treatments and therapeutic regimens, and consequently, in this section, the dynamics of methanogens in different diseased conditions will be described. In the future, it may be possible to use the dynamics of the methanogens to detect or diagnose conditions by quantifying methanogens in human biopsies or stool samples.

Obesity

Obesity is a chronic, metabolic, lifestyle health disorder of enormous public health importance (Rippe et al. 1998; Mathur and Barlow 2015). There is convincing evidence that H_2 removal by methanogens in the gut allows for more effective bacterial fermentation with more efficient production of short-chain fatty acids (SCFAs) (Samuel and Gordon 2006). These workers showed in mice that the presence of *M. smithii* correlates with increased SCFA levels which increases energy absorption in the lower intestines (Samuel and Gordon 2006). Studies have found a higher population density of *Methanobacteriales* in obese individuals when compared to their normal-weight counterparts. Interestingly, a decrease in the number of *Methanobacteriales* was observed in these persons after weight-reducing surgeries (DiBaise et al. 2008). Thus, methanogen population dynamics could serve as a reliable marker for obesity pathogenesis. However, a more conclusive relationship remains to be established between genetics, diabetes, obesity, and methanogens. Significantly higher numbers of H_2-utilizing methanogens were observed in obese individuals than in normal-weight or post-gastric-bypass individuals leading to the hypothesis that interspecies H_2 transfer between bacterial and archaeal species increases energy uptake by the human large intestine in obese persons (Million et al. 2016). However, in obese individuals, methanogens would increase calorie uptake by supporting the growth of fermentative bacteria and could exaggerate the obesity further thus leading to severe health conditions.

Anorexia

In contrast to the higher levels of methanogens in obese individuals, it would be expected that low methanogen populations are indicative of possible anorexia. However, *M. smithii* populations are higher in individuals that suffer from anorexia nervosa than in healthy individuals (Armougom et al. 2009). Surprisingly, the methanogen populations in anorexic subjects were found to be even higher than obese individuals (Armougom et al. 2009). For anorexic patients, the increase in *M. smithii* would lead to increased energy efficiency in severely low calorie diets. Anorexic patients often suffer from constipation which is also associated with an increased production of methane and the associated higher numbers of methanogens (Fiedorek et al. 1990).

Colorectal cancer

Colorectal cancer, the third most common cancer in the world, begins in the colon or rectum of the large intestine. Several anaerobic microbes have been implicated in the development or prevention of colorectal cancer (Kontou et al. 2012). Study utilizing breath-methane as an indicator of bowel cancer risk in African populations found that there is no correlation between the presence of methanogens in stools and bowel cancer (Segal et al. 1988). Similar findings were obtained in another study where 45% of patients with colorectal cancer had detectable methanogens in comparison to 48% in the healthy controls. The dominant methanogen in these two groups was *M. smithii* (Scanlan et al. 2008). In contrast, one study indicates that 80% of colorectal cancer patients produce breath-methane, which was seen as a preliminary indication of methanogens (Haines et al. 1977). It is therefore apparent that methanogenic populations could be useful markers for colorectal cancer detection, but further investigation is required.

Inflammatory bowel disease

Inflammatory bowel disease (IBD) is a collective term for inflammatory diseases of the bowel, namely, ulcerative colitis (UC) and Crohn's disease (CD) (Nasseri-Moghaddam 2012). The gut microbiota is associated with the modulation of mucosal immunity and the etiology of IBDs. Convincing evidence is accumulating that links methanogenic populations of the gut to inflammation (Barbara et al. 2005; Pimentel et al. 2012). Initial studies found that there is a correlation between methane excretion and IBD. In UC and CD patients, a mere 10 to 30% were methane producers, compared to 50% in control groups. These findings have been supported using molecular profiling (Pimentel et al. 2012). In another study, it was reported that 24% of UC and 30% CD patients had detectable levels of methanogens while in the control group of healthy individuals, 48% of the individuals had detectable methanogens (Scanlan et al. 2008). Recent evidence supports that two main archaeal species found in the digestive system of humans, *Methanobrevibacter smithii* and *Methanosphaera stadtmanae*, can have differential immunogenic properties in the lungs of mice, with *Methanosphaera stadtmanae* being a strong inducer of the inflammatory response (Table 1) (Lecours et al. 2014). One possible reason for a decrease in methanogen populations in IBD patients could be due to diarrhea removing the communities from the intestinal tract—effectively "colonic purging" of methanogens (Pimentel et al. 2003).

Irritable bowel syndrome

IBS is a common gastrointestinal disorder for which various causes are suggested. Unlike IBD, IBS is not associated with inflammation or chronic damage to the bowel (Bajaj et al. 2012). Patients suffering from IBS can be split into two main subgroups: those that have diarrhea-predominant symptoms (IBS-D) and those with constipation-dominant symptoms (IBS-C). Recent studies have made links between gut microbiota disturbances and symptoms of IBS, including bloating and constipation. Vast differences have been noticed with breath-methane excretion levels in IBS-C and IBS-D patients compared with previous studies. It has been shown that the proportion of methane producers is higher in IBS-C patients (58% methane producers) than in IBS-D patients (28% methane producers) (Table 1) (Triantafyllou et al. 2014). This finding is consistent with the report of higher levels of methanogens in anorexia patients who often had constipation. This is indicative of a possible link between higher levels of methane and slower gut transit times. However, this has not been shown for IBS patients relative to healthy controls since the IBS patients have not been divided into IBS-D and IBS-C. For example, in another study, researchers noted that in the healthy control group, 15% were methane producers compared to 20% in the IBS group. Interestingly, the IBS group that was methane-positive was more likely to suffer from constipation (Bratten et al. 2008). One study using molecular techniques for profiling the microbiota also found a lack of correlation between IBS and methanogen content with 48% of both the control group and the IBS group classified as methane producers (Scanlan et al. 2008). It may be possible that within the IBS patients, methane production and methanogenic archaea presence may be used to subdivide the subjects.

Diverticulosis

Diverticulosis is a condition affecting the large intestine. Complications associated with diverticulosis are a major concerns in the USA and throughout the western world (Yazici et al. 2016). Some researchers examined the correlation between colonic gases, such as methane, with the development of diverticulosis in humans. Patients suffering from this ailment develop diverticula, or pouches, in their intestinal wall. Diverticulosis is commonly undiagnosed due to generic symptoms, such as gut pains, that can be attributed to other diseases. In 1986, a group of researchers studied methanogen populations in patients undergoing sigmoidoscopy (Weaver et al. 1986). The study utilized microbial enumeration, cell culturing, and methane production to determine how many methanogens were present per gram of dry stool (gds) (Miller et al. 1982; Weaver et al. 1986). Samples were divided into three groups depending on methanogen content: high (1×10^7 methanogens/gds), low (less than 1×10^7 methanogens/gds), and absent (0 methanogens/gds). A significant difference was seen in diverticulosis patients in comparison to normal controls in relation to the "high" group. Only 25% of the healthy control group belonged to this category, whereas 58% of patients suffering from diverticulosis contained more than 10^7 methanogens/gds. There were no significant differences detected in the low and absent groups. It can be suggested that high concentrations of methanogens may be used as a health marker when diagnosing diverticulosis (Weaver et al. 1986).

Another study conducted in the USA showed that individuals having detectable methane in their breath (> 5 ppm) were suffering from diverticulosis as compared to those who did not have detectable levels of less methane (below the threshold level of 5 ppm) in their breath. Consequently, these workers proposed an association between methanogens and diverticulosis (Yazici et al. 2016).

Atherosclerosis

Atherosclerosis is a cardiovascular disease caused by chronic inflammation in large and medium-sized arteries (Goodman and Gordon 2010; Loscalzo 2011). There are various risk factors associated with atherosclerosis including advanced age, hypertension, and red meat consumption. Historically, the trans-fatty acids were considered the main link between the disease and red meat, however, recent studies have identified other components which may be involved. Choline and L-carnitine are abundant in red meat, and are converted into trimethylamine (TMA) by the intestinal microbiota (Polychronopoulos et al. 2010; Liu et al. 2015). TMA is absorbed by the intestine and transported to the liver, where it is further metabolized by flavin monooxygenase (FMO) to trimethylamine N-oxide (TMAO). TMAO has been linked to atherosclerosis (Brugère et al. 2014). A novel approach to treating atherosclerosis could be to us archaea as live supplements, archaebiotics. This concept refers to the use of methylotrophic methanogens, such as *Methanomassiliicoccus luminyensis*, to reduce the amount of TMA being absorbed by the intestines (Hippe et al. 1979; Brugère et al. 2014). If successful, the administered methylotrophic methanogen would reduce TMA into methane which would be excreted in the breath or as flatulence. Based on this theoretical concept, it can be suggested that people who naturally contain high levels of *Methanomassiliicoccus luminyensis* in their intestines are less susceptible to atherosclerosis (Brugère et al. 2014).

Periodontitis

Periodontitis is a disease of the oral cavity caused by overgrowth and infection by multiple microorganisms. Periodontitis affects around one third of the population, and is a known precursor for loss of teeth and endocarditis (Li et al. 2000). A link between the presence of archaea, and in particular methanogens, and the disease has been proposed (Lepp et al. 2004). Molecular methods have been used in these studies, primarily FISH and RT-qPCR, which give an accurate and reliable insight into the microbial community structures in this diseased state. There are numerous reports that methanogenic archaea are present at higher levels in the oral cavity of subjects with periodontitis than in healthy subjects (Lepp et al. 2004; Vianna et al. 2008; Li et al. 2009). In another study, researchers found that in patients with severe periodontitis, 18% of the microbial population consisted of archaea, which was equal to 1×10^6 gene copies per microliter. In comparison, archaeal populations were not detected in the healthy controls (Lepp et al. 2004). Additionally, a study by another group of scientists reported that the prevalence of archaea in the oral cavity of those with periodontitis was reduced by treating the disease (Lira et al. 2013).

Conclusions

In the last decade, the development of the metagenomics, advanced bioinformatics, and statistical tools has provided an enormous amount of evidence about the role of gut microbes in human health and how shifts in their relative abundance have been associated with various diseases. Recent diversity analysis-based studies showed an increased diversity of the archaeal component of human gut microbiota than was previously demonstrated. With the application of next generation sequencing technologies, more direct links with non-communicable diseases and methanogen counts may be achievable. Most of the older studies that examined the effect of additives on the structure and function of microbiome, employed anaerobic culture based techniques, and results were influenced by the stringency of the anaerobic methodology used. Since the sequencing technologies are not reliant on stringent anaerobic culturing of the archaea, one can expect that more reliable comparisons can be made between results from different laboratories.

Because of their unique metabolism (ability to metabolize especially trimethylamines and other methyl compounds by members of a newly discovered order of methanogen, i.e., *Methanomassillicoccales*), methanogen populations could play an important role in the progression of some diseases. Several studies in the last decade directly or indirectly suggested a link between methanogens and a number of metabolic diseases. In addition, because of the role of methanogens as a hydrogen sink in the gut, methanogens should be considered as an important target for the studies relating to modulation of the gut microbiota in humans for improving health or reducing risk of disease. Consequently, there is merit in proposing dietary intervention studies targeting the manipulation of archaeal components of the gut microbiome, thereby influencing the methanogen populations for improved health. This concept could even be extended to consider their use as a probiotic whereby live preparations of archaea could be used as dietary supplements (archaeobiotic). It may be feasible to manipulate the methanogens for a therapeutic or prophylactic benefit for health.

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Compliance with ethical standards

This article does not contain any studies with human participants or animals performed by any of the authors.

Conflict of interest The authors declare that they have no conflict of interest.

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